Cayenne Clusters Muesli Project Presentation

Muesli Delivery EDA

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Agenda

01 Data: Overview & Cleaning

02 Delivery Process: Flow & Steps

03 Our Approach

04 Main KPIs Explained

05 Diagnostic analytics

06 Conclusions & Recommendations

01 Data Overview and Cleaning

Orders

Total records: 5009
Unique orders: 5009

Main info:

- order_date
- ship_mode
- origin_channel

Interns

Total records: 204
Unique orders: 204

Main info:

- ready_to_ship_date
- pickup_date

Process

Total records: 3002
Unique orders: 3002

Main info:

- on_truck_scan_date
- truck_ship_mode

Campaign

Total records: 333
Unique orders: 333

Main info:

- arrival_scan-date

01 Data Overview & Cleaning

Key Takeaways:

- The project uses **several datasets**: orders, intern warehouse observations, shipment scans, and customer arrival scans
- The data was cleaned to make sure column names and date formats were **consistent**
- Data columns were converted so we could **calculate time differences** between delivery stages
- After cleaning, the datasets were **merged** to reconstruct the full delivery process for each order

02 Delivery Process: Flow & Steps

Data Sources

Orders
order_date
ship_mode

Intern Warehouse Observations
ready_to_ship_date
pick_up_date

Shipment Scan Data
on_truck_scan_date
truck_ship_mode

Customer Arrival Scan Data
arrival_scan_date

Order

Customer places an order

Order is recorded in the system

Warehouse Processing

Order is processed in the warehouse

Processing time: ~2 days

Truck Picks Up

Logistics company picks up the order

Trucks leave: Monday, Wednesday, Friday

Transportation

The order is transported to the customer

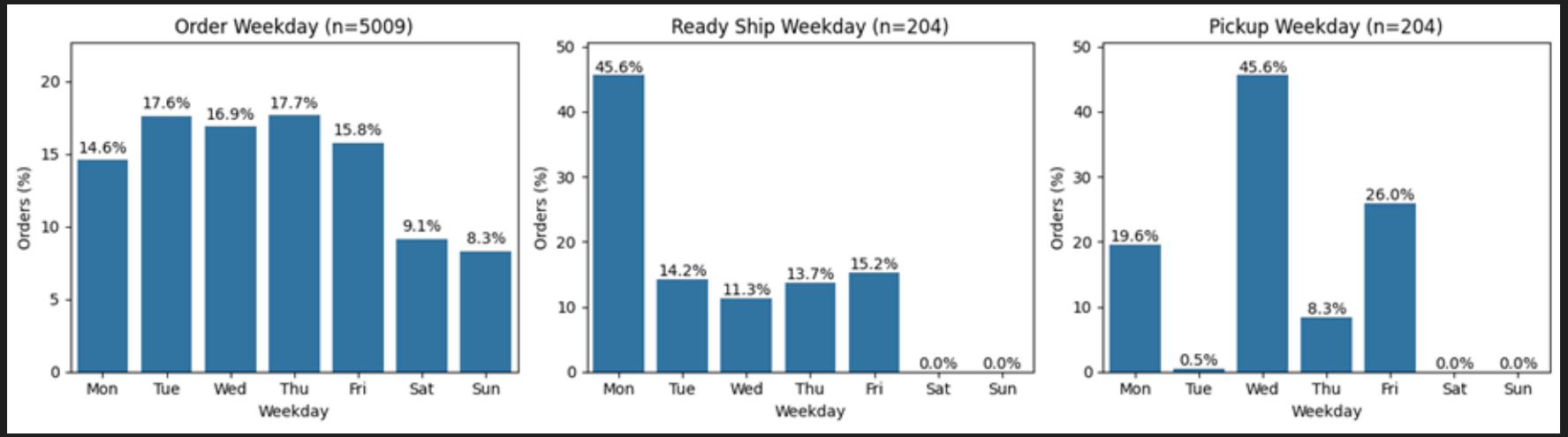
Average delivery time: ~3 days

Arrival

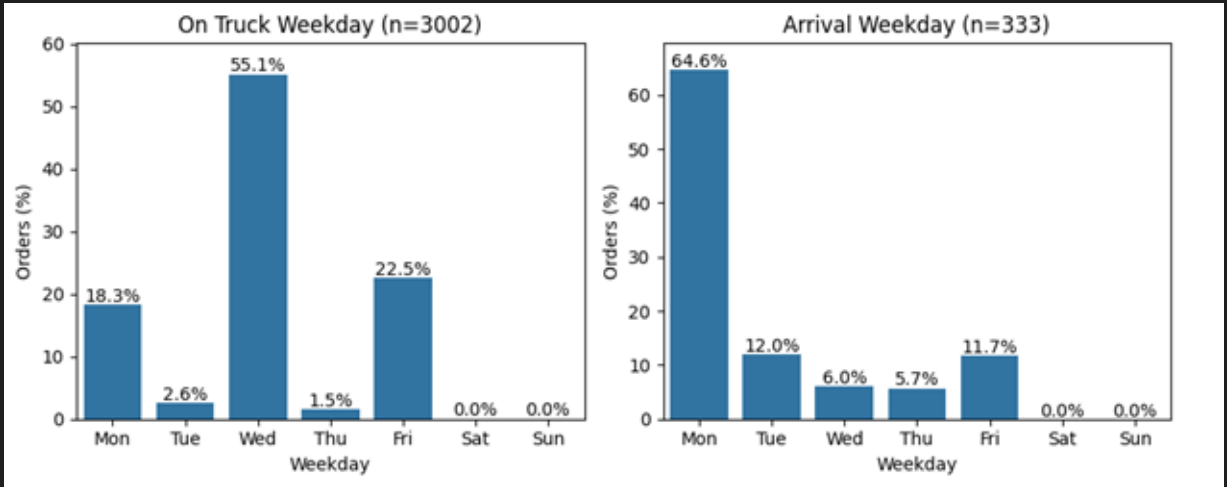
Customer receives the order

Arrival scan recorded via QR code

Order ››› Warehouse ››› PickUp ›››



You Order Is on the Way >>> Delivery!



Our Approach:

Calendar Days vs. Business Days

In our group we used **both** approaches to calculating the main metrics:

Calendar Days

"Pessimistic" Assumption

clients make orders all week

Business Days

"Optimistic" Assumption

warehouse doesn't work on weekends;
trucks pick up orders on Mon, Wed, Fri only

04 Main KPIs Explained

Stages	From	To	Data Available
Warehouse Processing	Order date	Ready to ship day	Partial (intern data)
Waiting for Truck	Ready to ship date	On truck scan date	Partial (intern data)
Full Warehouse time	Order date	On truck scan date	Partial (truck company data)
Delivery	On truck scan date	Arrival scan date	Partial (truck company data + campaign data)
Total Delivery Time	Order Date	Arrival Scan Date	Partial (campaign data)

04 Warehouse Processing Time

Ready to ship date (Intern data) - Order date (Order process data)

Scenario 1
(with duplicates, calendar days):

* n = 753
* **mean = 4.3 days**
* median = 5 days
* max = 8 days

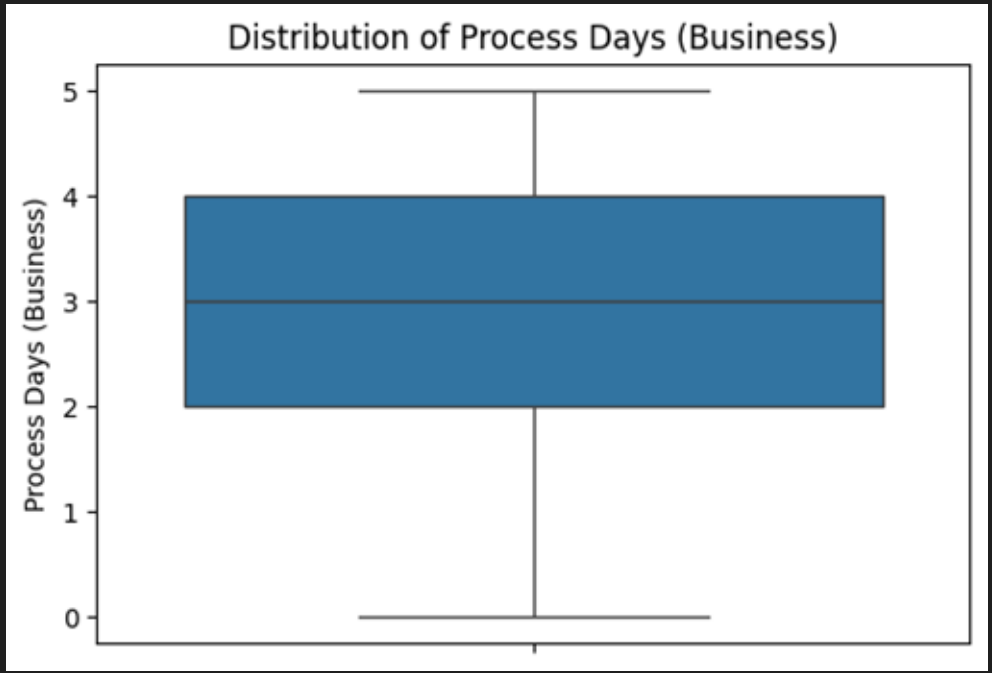
Company claims~2 days



04 Warehouse Processing Time

Scenario 2 (business days)

Warehouse claims that the processing of each order takes on **average 2 days.**



Business Days Stats:

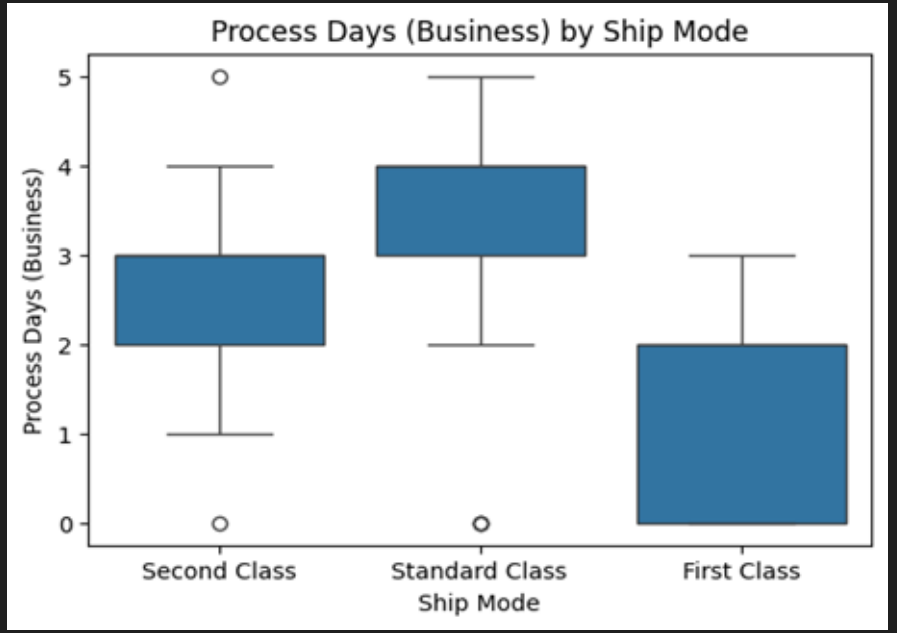
mean	**2.769608**
std	1.417991
min	0.000000
25%	2.000000
50%	**3.000000**
75%	4.000000
max	5.000000

Ready to ship date - Order date
n = 204

04 Warehouse Processing Time
by shipment mode

Business Days Stats:
(for the entire sample)

mean	**2.769608**
std	1.417991
min	0.000000
25%	2.000000
50%	**3.000000**
75%	4.000000
max	5.000000



Ready to ship date - Order date
n = 204

04 Waiting for Truck

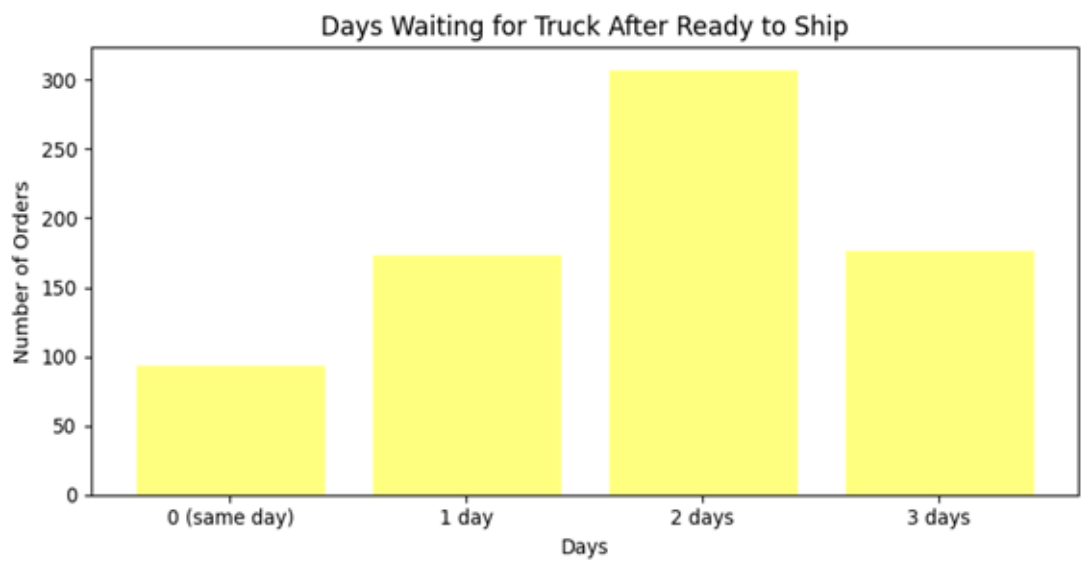
Pick up date (Intern data) - Ready to ship date (Intern data)

Scenario 1
(with duplicates, calendar days):

* n = 753
*** mean = 1.75 days**
* median = 2 days
* max = 3 days

Company claims = 1 day

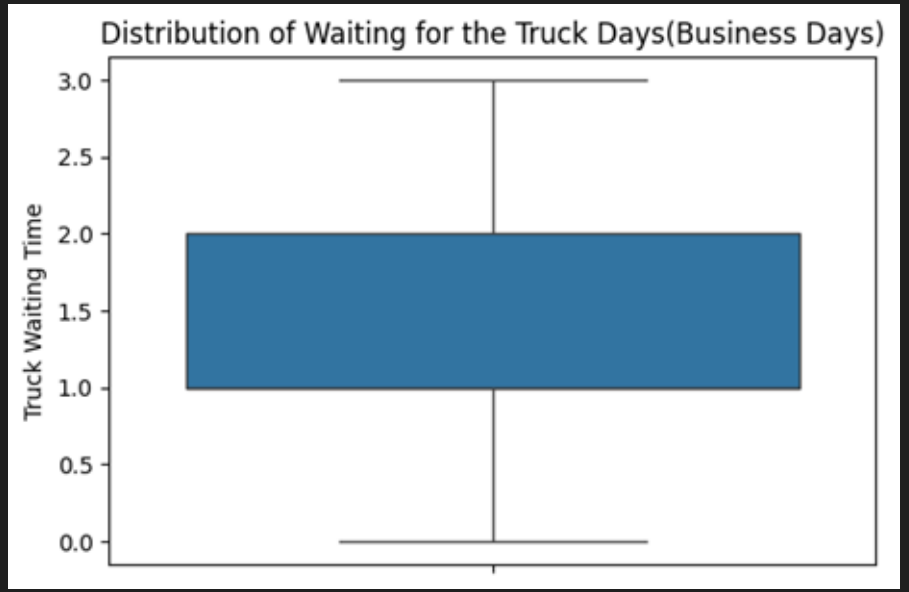
Same days pickups = 12.5%



04 Waiting for Truck Time

Scenario 2: No duplicates, business days

Truck company claims that the truck pick up timeline is **one day on average**



Business Days Stats:

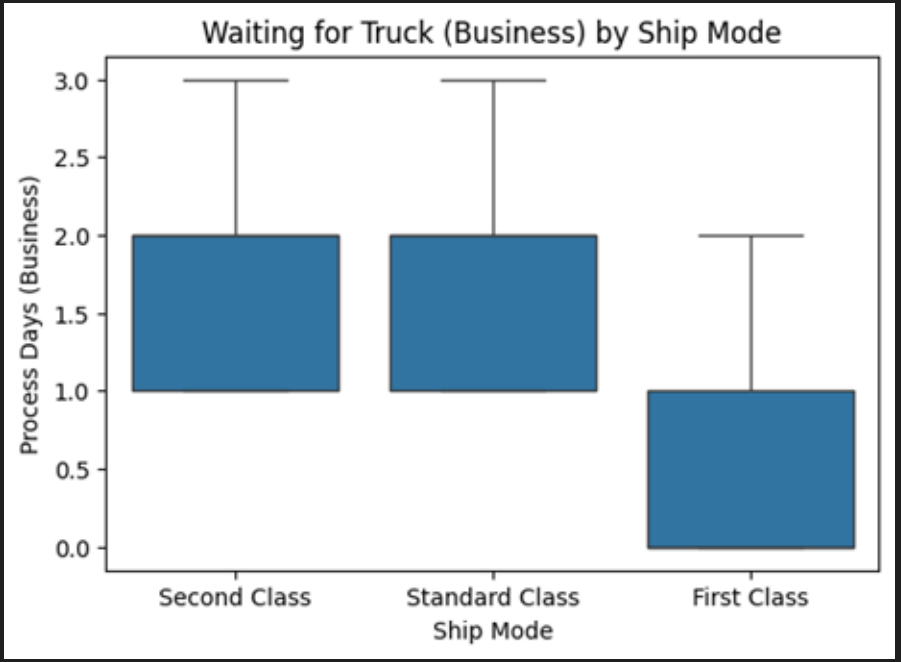
mean	**1.357843**
std	0.821141
min	0.000000
25%	1.000000
50%	**1.000000**
75%	2.000000
max	3.000000

Pick_up_date - ready_to_ship_date
n = 204

04 Waiting for Truck Time
by shipment mode

Scenario 2:
No duplicates, business days

Truck company claims that the truck pick up timeline is **one day on average.**




```
Pick_up_date - ready_to_ship_date
n = 204
```

04 Full Warehouse Time

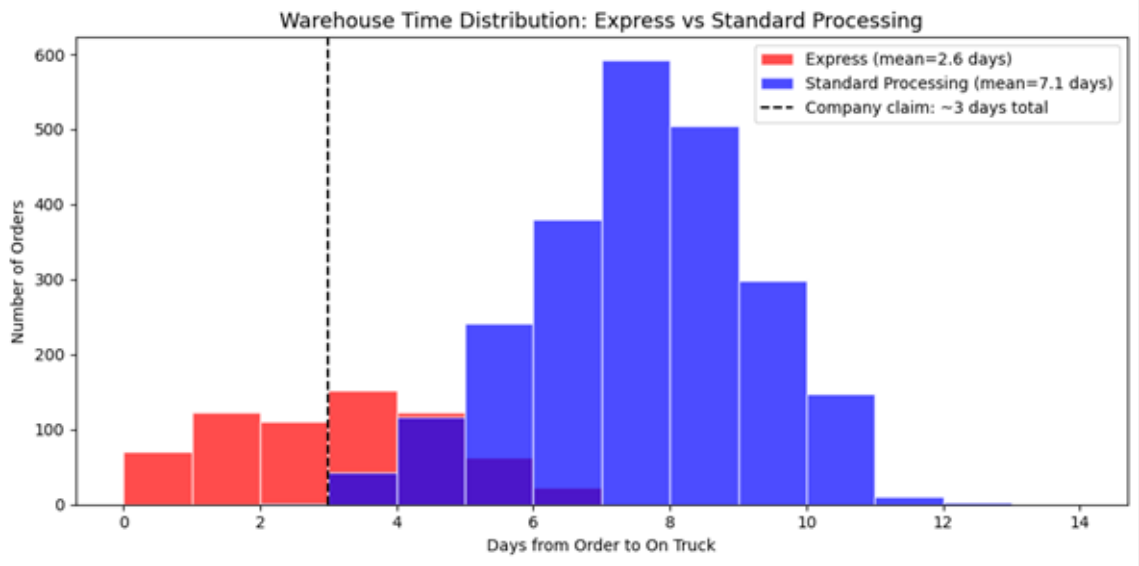
On truck scan date (Order process data) - Order date (Order process data)

Scenario 1
(with duplicates, calendar days):

* n = 5899
*** mean = 6.1 days**
* median = 7 days
* max = 12 days

Company claims = 3-4 days

- Express mean = **2.6 days**
- Standard processing mean = **7.1 days**



04 Full Warehouse Time: warehouse vs. truck company

We can compare the small data from warehouse (interns) to the big data (from the truck company)

Warehouse Data:

count	204.000000
mean	**4.127451**
std	1.839209
min	0.000000
25%	3.000000
50%	**5.000000**
75%	5.000000
max	7.000000

[Pick_up_date - order_date], Business days

Truck Company :

count	3002.000000
mean	**4.331779**
std	1.847353
min	0.000000
25%	3.000000
50%	**5.000000**
75%	6.000000
max	9.000000

[On_truc_scan_date - order_date], Business days

Good

Two datasets match >>> we can trust the assumption about the average amount of days for warehouse processing

Bad

It's higher than the promise from the warehouse

04 Delivery (Transportation) Time

Arrival scan date (Campaign data) - On truck scan date (Order process data)

Scenario 1
(with duplicates, calendar days):

* n= 668
* **mean= 4.6 days**
* median= 5 days
* max= 7 days

Company claims= **3 days**



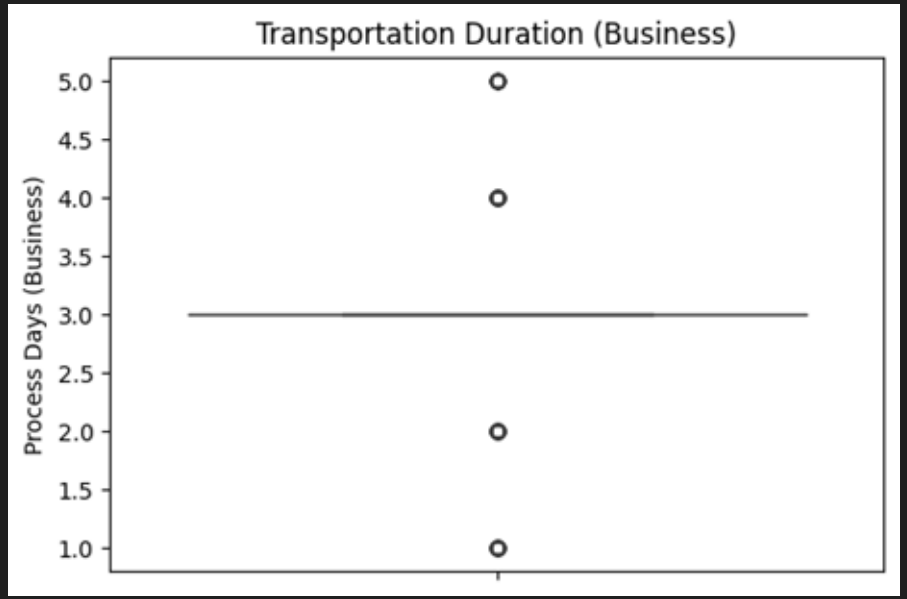
04 Delivery (Transportation) time

Arrival_scan - On_truck_scan
n = 333

Scenario 2: No duplicates, business days

Truck company claims that the truck pick up timeline is **three days on average.**

Haha. Some pretty confusing boxplot, right?



Business Days Stats:

count	333.000000
mean	**2.921922**
std	0.956855
min	1.000000
25%	3.000000
50%	**3.000000**
75%	3.000000
max	5.000000

04 Total Delivery Time

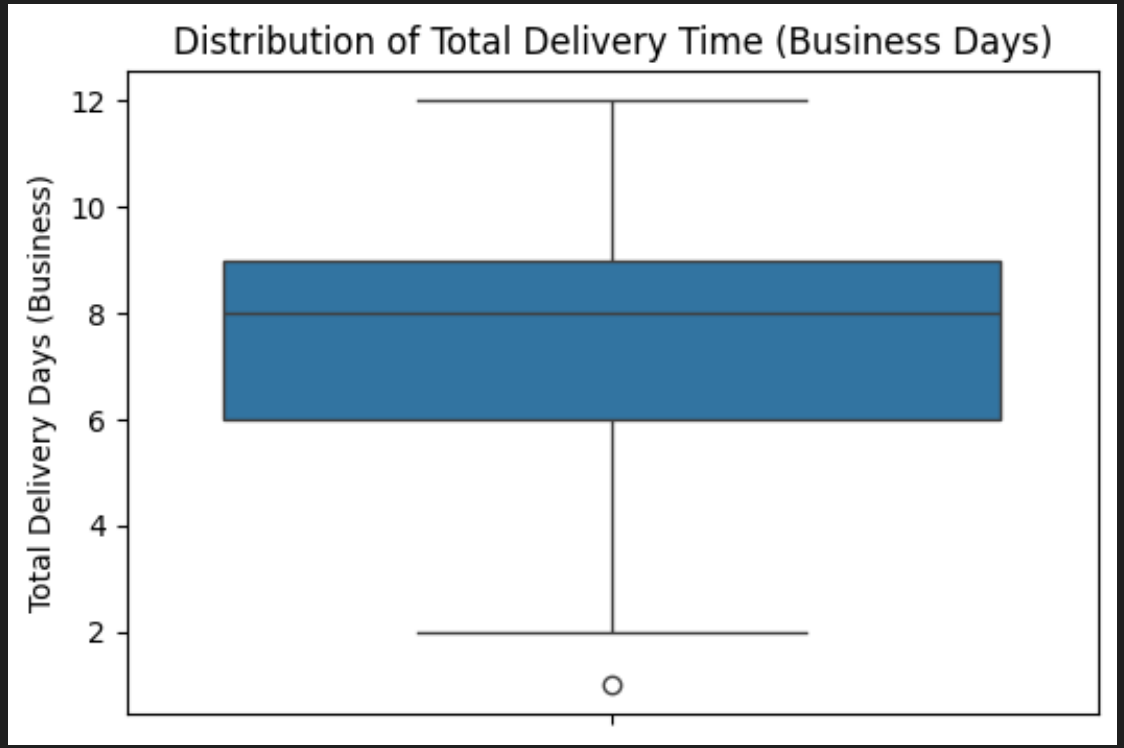
Arrival_scan - Order_date
n = 333

Muesli Company claims that all the orders are normally take **7 days on average** to be delivered.

Scenario 2: No duplicates, business days

```
Business Days Stats:

count    333.000000
mean       7.399399
std        2.050477
min        1.000000
25%        6.000000
50%        8.000000
75%        9.000000
max       12.000000
```

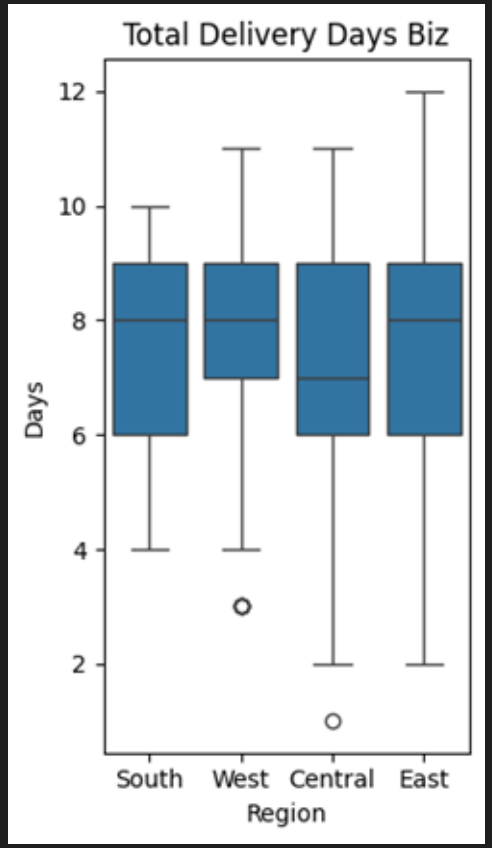


04 Total Delivery Time: by region

Arrival_scan - Order_date
n = 333

The region (where to the order is supposed to travel) **does not affect** much the total delivery time.

Scenario 2: No duplicates, business days



Business Days Stats:

	Central	East	South
		West	
count	77	92	
	49	115	
mean	**6.9**	**7.34**	
	7.63	**7.68**	
std	2.15	2.09	
	1.79	2.01	
min	1	2	
	4	3	

04 Total Delivery Time: by shipment mode*

```
Arrival_scan - Order_date
n = 333
```

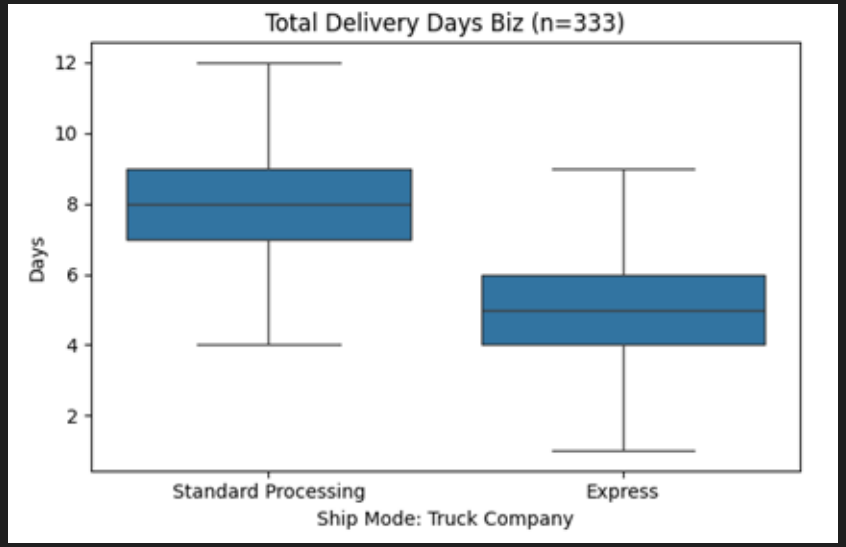
* Truck Company and Muesli Company use **two different approaches** to the shipment mode:

Muesli company has **three options** (First Class, Second Class, Standard Class) and Truck company has **two options** (Express and Standard Processing)

Scenario 2: No duplicates, business days

Business Days Stats:

	Express	Standard Processing (First+Second Class)
count	75	258
mean	**4.8**	**8.2**
std	1.52	1.49
min	1	4
50%	**5**	**8**
max	9	12



04 KPIs summary

Stages	Company claim	Average (Scenario 1)*	Average (Scenario 2)**
Warehouse processing	~2 days	4.3 days	2.8 days
Waiting for truck	1 day	1.8 days	~1.36 days
Full warehouse time	3-4 days	6.1 days	~4 days
Delivery	~3 days	4.6 days	2.9 days
Total delivery time	9-10 days	-	**7.4*** days**

* Scenario 1 takes into account **calendar** days

** Scenario 2 calculates the averages based on **business** days

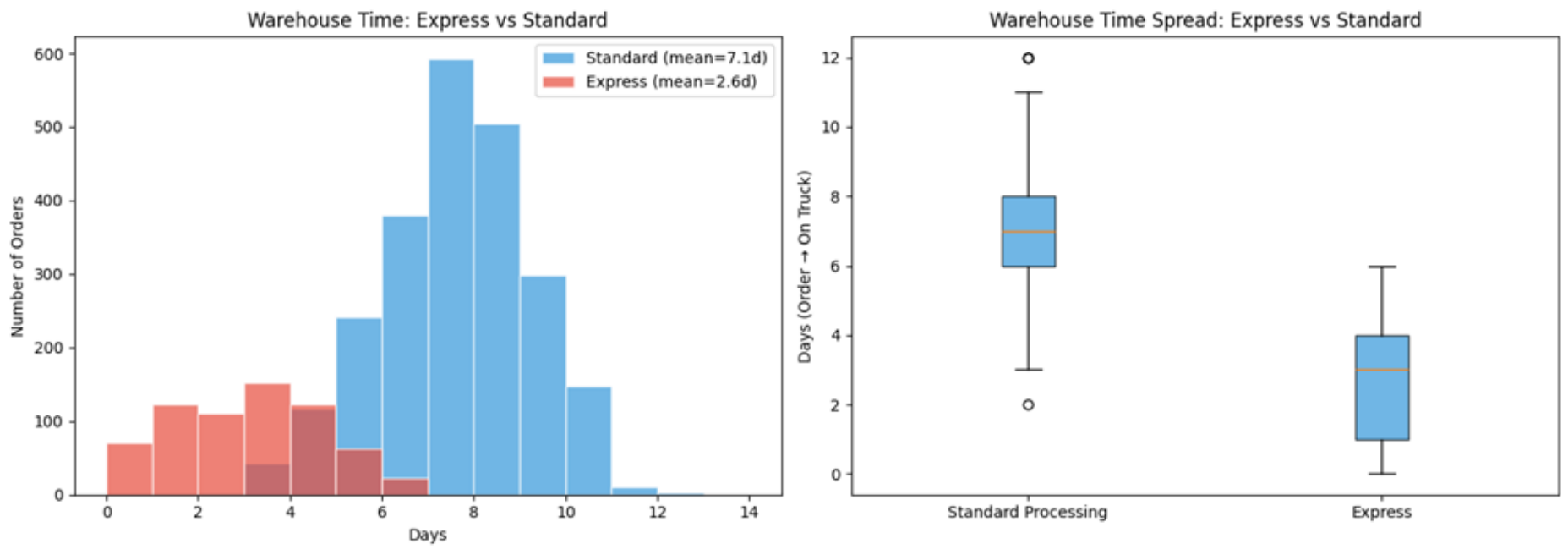
***** Total delivery time varies significantly by the shipment mode: 4.8 vs. 8.2!!**

05 Diagnostic analytics – Where are the problems?

- **Express vs standard** - Is express worth it?
- Does **day of order** affect warehouse time?
- Does it matter, **when to place the order**?
- Delivery reliability - How **consistent** is the logistics company?

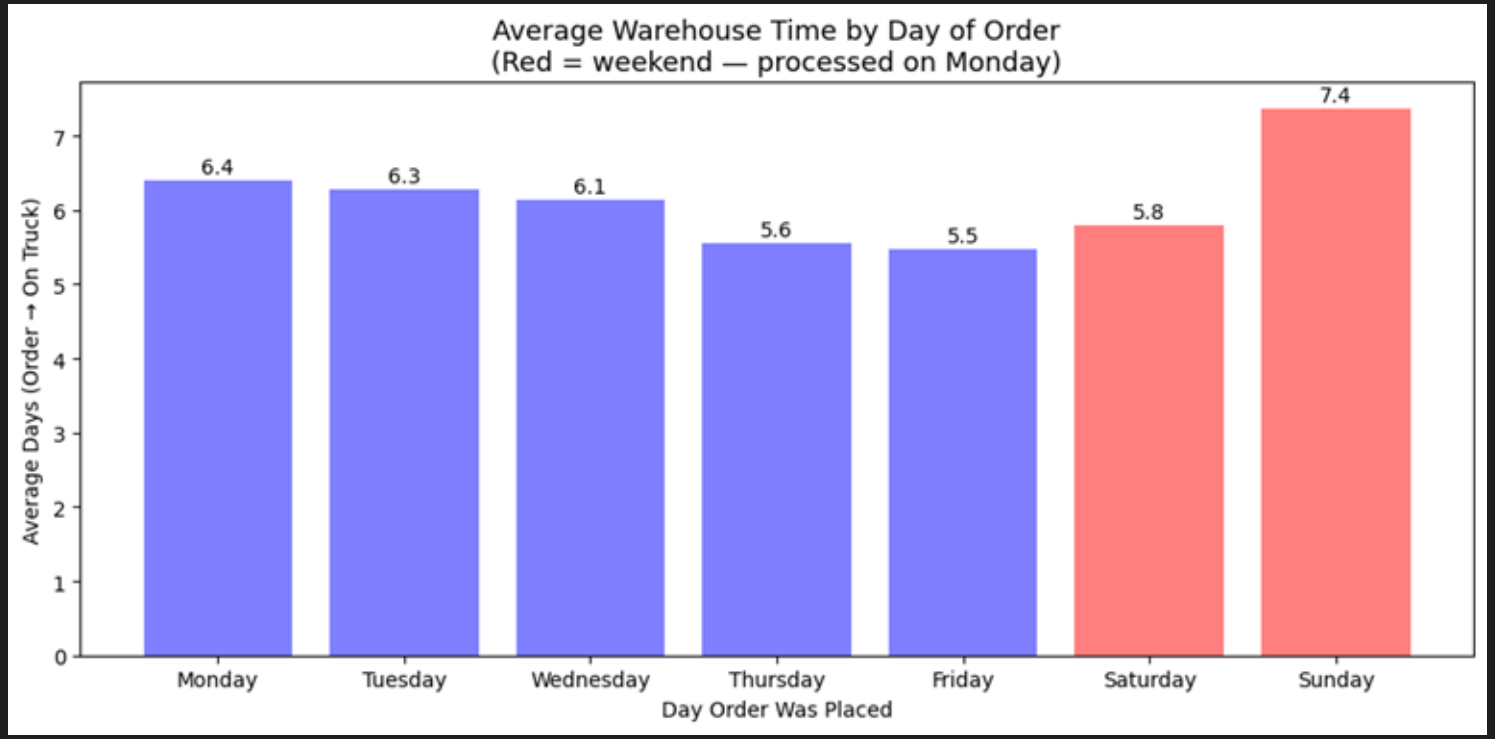
05 Diagnostic analytics – Where are the problems?

Express saves on average 4.5 days in warehouse time.



05 Diagnostic analytics - Where are the problems?

Weekend orders should take longer since the warehouse only operates Mon–Fri.



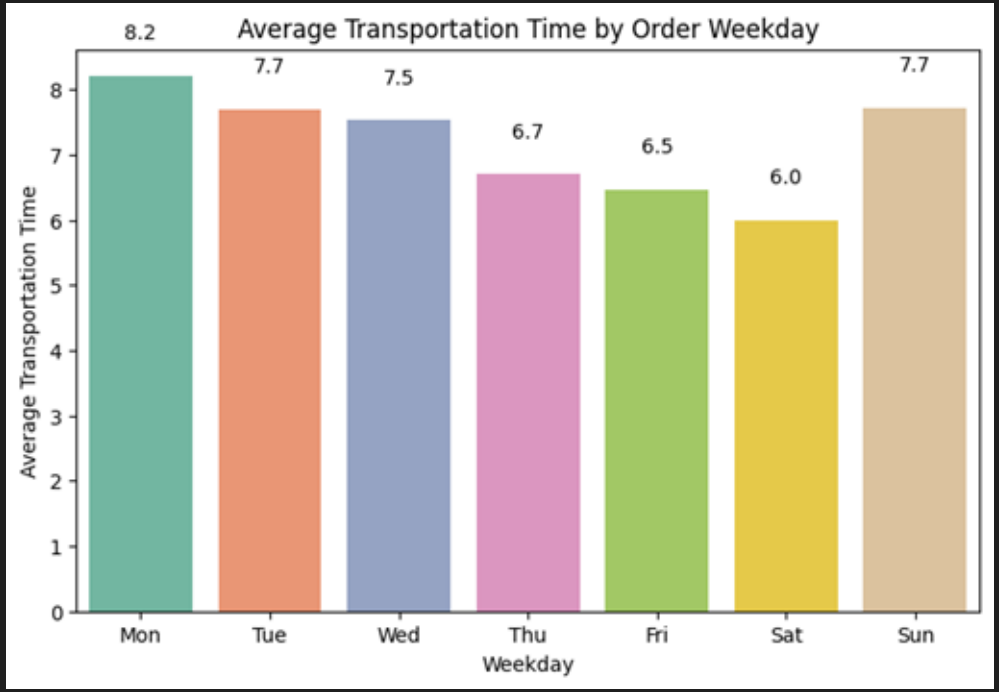
05 Diagnostic analytics – Where are the problems?

Order date vs. Total delivery

Does it matter, when a client place their order? Apparently, **yes!**

The graph shows how many days the delivery is going to take depending on the order day.

If you ordered a product on Monday, it will be travelling appx. for **8.2 days.**



05 Diagnostic analytics – Where are the problems?

Delivery reliability - Is the logistics company consistent enough?





CONCLUSIONS: Scenario 1

1/ Warehouse is the biggest bottleneck.

Processing takes **more than double the claimed 2 days.**

2/ Logistics company is also underperforming.

Delivery takes **4.6 days on average**. Only ~21% of orders arrive within the promised 3 days.

3/ Express processing is effective.

It **reduces total warehouse time**. Worth investigating, if it could be offered more broadly.

4/ Weekend orders add delay

~15% of orders are placed on the weekends.
Naturally takes longer since the warehouse is closed.

CONCLUSIONS: Scenario 2

1/ Warehouse processes too many orders on Mondays.

Only **25% of all orders** are processed in 2 days or less.

2/ Overall performance of the truck company is satisfactory.

But only **First Class** deliveries are picked up in the promised range of time.

3/ Information about truck arrivals is inconsistent.

Both warehouse interns and truck company own scan data shows that trucks perform pick ups on **Tue and Thu**.

4/ Average delivery time is about 7 days.

But the difference between shipment modes is almost doubled:
4.8 for express and **8.2 for standard** delivery.

RECOMMENDATIONS

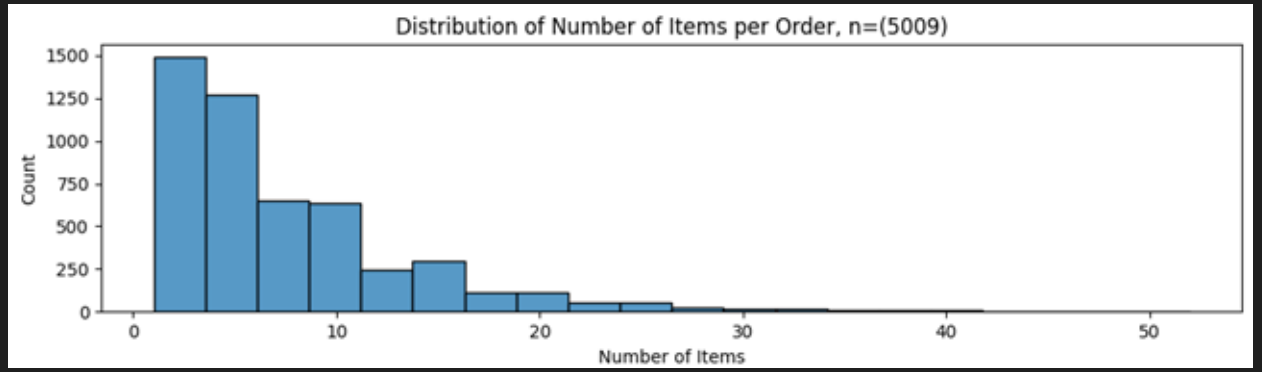
KPIs to track going forward...

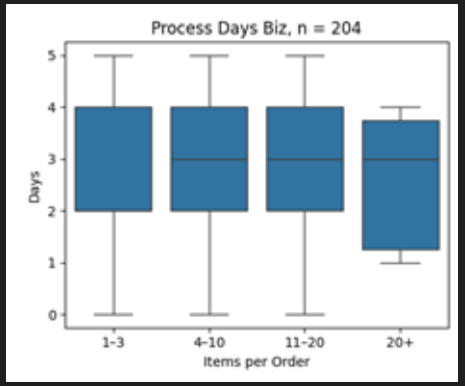
KPI	Target
Average warehouse processing time	≤ 2 days
% orders processed within 2 days	≥ 80%
Average delivery time	≤ 3 days
% orders delivered within 3 days	≥ 80%
Express vs standard time saving	Monitor monthly

Thank you!

Contact our team

Size of an Order: doesn't matter

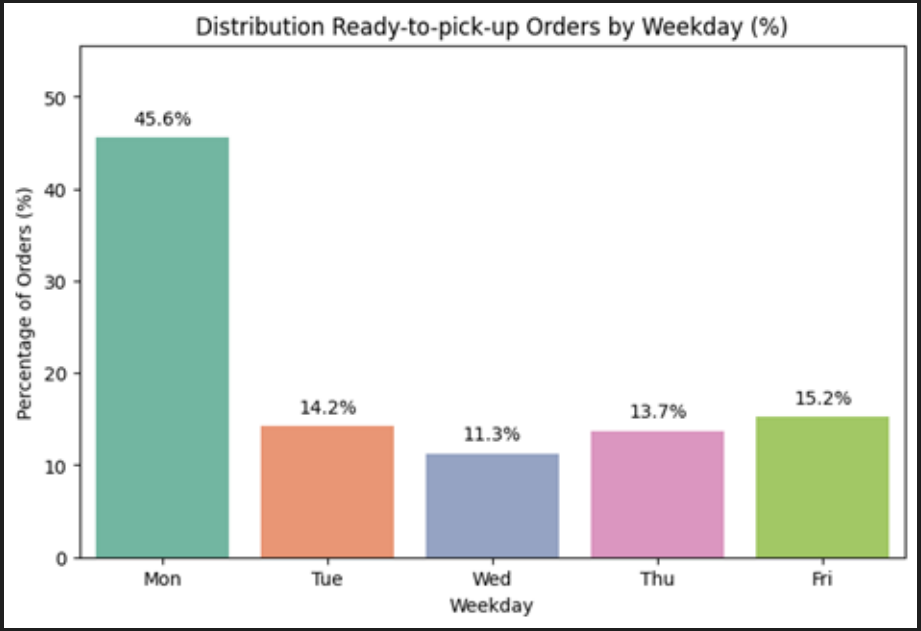




There are many small orders (below 10) and very few pretty big orders (probably, just outliers), and we wanted to check if the size of the order affects the warehouse processing time.

Nope! It doesn't really.

Do We Need More Truck Days?
maybe



The most orders are being picked up on Monday (45.6% of **all orders**).

It would be nice either to process weekend orders on Sat or Sunday

OR

Have pick ups on Tuesday as well.